

Code No: 157BC**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech IV Year I Semester Examinations, July/August - 2023****DATA MINING****(Common to CSE, IT, ITE)****Time: 3 Hours****Max. Marks: 75**

- Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) Why do we need data preprocessing? [2]
- b) What are major issues in data mining? [3]
- c) What is SPM? [2]
- d) List and explain the measures required for mining interesting association rules. [3]
- e) What is bayes theorem? [2]
- f) What is tree pruning? [3]
- g) What is manhattan distance measure? [2]
- h) When do you say an object is density reachable? [3]
- i) What is multimedia data? [2]
- j) How text mining is performed? [3]

PART – B**(50 Marks)**

2. What are various kinds of knowledge that can be extracted by data mining? Explain them. [10]

OR

3. What is noisy data? Explain how it is handled in data preprocessing stage. [10]

4. What is correlation analysis? Why is it required? What are various correlation measures? Explain them in detail. [10]

OR

- 5.a) What are multilevel association rules? What are different types of multidimensional association rules? Explain them.

- b) What are inter and intra dimensional association rules? Explain them. [5+5]

6. What is decision tree? Write down decision tree induction algorithm. Explain various attribute selection measure for decision tree induction. [10]

OR

7. What is rule based classification? What are various rule assessing measures? How conflict resolution is done in rule based classification? [10]

8. Explain STING clustering algorithm. [10]
- OR**
9. Explain DENCLUE algorithm for clustering. [10]
10. Explain how mining can be performed on stream data. [10]
- OR**
11. Write a short note on mining World Wide Web. [10]

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JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B.Tech IV Year I Semester Examinations, December-2023/January-2024

DATA MINING

(Common to CSE, IT, ECM, ITE)

Time: 3 Hours

Max. Marks: 75

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PART – A

(25 Marks)

- 1.a) What is a data warehouse? [2]
- b) How does constraint-based method influence the discovery of association rules? [3]
- c) What is the advantage of rule-based classification systems? [2]
- d) List the major clustering methods. [3]
- e) Define time-series data mining. [2]
- f) What is the purpose of data preprocessing in data mining? [3]
- g) What is association rule mining? [2]
- h) How does a Bayesian classifier handle uncertainty? [3]
- i) Mention the applications where clustering is commonly used. [2]
- j) Briefly describe multimedia data mining. [3]

PART – B

(50 Marks)

- 2.a) How do task primitives contribute to the overall effectiveness of data mining? Explain.
b) Explain the significance of data preprocessing in data mining. [5+5]
- OR**
- 3.a) Compare data mining with data warehouse.
b) Provide a step-by-step explanation of how data mining can be applied to classify data. [5+5]
4. Explore the different mining methods used in association rule mining and discuss their advantages and limitations. [10]
- OR**
- 5.a) Explain the process of mining frequent patterns.
b) Describe the role of correlation analysis in identifying relationships within datasets.[5+5]
6. Provide a step-by-step explanation of how a lazy learning algorithm, such as k-Nearest Neighbors (k-NN), makes predictions. [10]
- OR**
7. Compare and contrast rule-based classification systems with decision tree induction. [10]

- 8.a) Discuss the relationship between density-based and grid-based clustering methods.
b) Describe the steps involved in a grid-based clustering approach. [5+5]

OR

9. How do partitioning methods work in cluster analysis, and what factors should be considered when choosing an appropriate method? Explain. [10]
10. Explain the following terms.
a) Mining the World Wide Web.
b) Spatial data mining. [5+5]

OR

11. Elaborate on the process of mining sequence patterns in transactional databases, outlining the steps and potential challenges involved. [10]

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PART – A**(25 Marks)**

- 1.a) What is meant by loose coupling of data mining system with data warehouse? [2]
- b) What is necessity of outlier detection? [3]
- c) What are succinct constraints? [2]
- d) What is all confidence correlation measure? [3]
- e) Distinguish between classification and prediction. [2]
- f) What is rule accuracy? [3]
- g) What are interval scaled variables? [2]
- h) Distinguish between supervised and unsupervised learning. [3]
- i) What is stream data? [2]
- j) What are the challenges in mining spatial data? [3]

PART – B**(50 Marks)**

2. What are data mining functionalities? Explain them. [10]
- OR**
3. Explain briefly about various tasks in data preprocessing. [10]
4. Generate interesting association rules using apriori algorithm for given data (min.support = 40%, min. confidence = 65%): [10]

TID	ITEMS
T01	A,B,D,E,F
T02	B,C,G,H
T03	A,C,D,G,H
T04	C,D,H
T05	D,E,F
T06	B,D,G,H

OR

5. Explain about constraint base association rule mining. [10]

6. Write down algorithm for decision tree induction. Give an example for decision tree. [10]

OR

7. Explain any one lazy learner with example and write algorithm for it. [10]

8. Simulate kmedoids clustering on following data where $k=2$, and A and B are initial centre points: (after finding initial clusters, find at least one non medoid which could be a medoid point). [10]

Data object	X	Y
A	2	6
B	3	1
C	5	7
D	6	7
E	8	8

OR

9. Explain DBSCAN clustering method. [10]

10. Explain multimedia data mining. [10]

OR

11. Explain how sequence patterns are mined from transactional databases. [10]

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